

CELL BIOLOGY

Introduction to cell

Cell is a structural, functional and hereditary unit of organism consisting of a small mass of protoplasm with nucleus or nuclear material which is surrounded by the plasma membrane with or without cell wall.

The term cell was given by Robert Hooke. He called Father of cytology.

The study of structure and composition of cell and components with the help of microscope is called cytology.

Swanson is called father of modern cytology.

Cell types: Prokaryotic and Eukaryotic cells

Prokaryotic cells are those cells that consist of one envelope system of organization but lack an organized nucleus or well-defined nucleus.

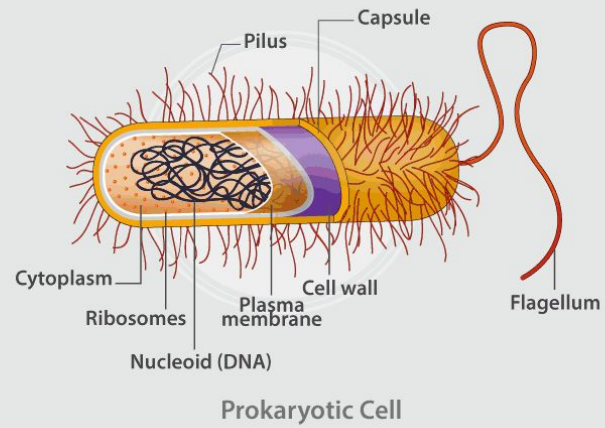
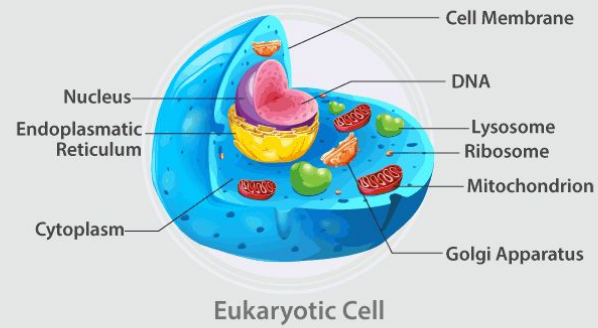
Primitive nucleus is present in cell without nuclear membrane, nucleoplasm and nucleolus but only chromosomes are present which are freely suspended in cytoplasm.

This type of nucleus is called as incipient nucleus, nucleoid.

	Prokaryotes	Eukaryotes
Type of Cell	Always unicellular	Unicellular and multi-cellular
Cell size	Ranges in size from 0.2 μm – 2.0 μm in diameter	Size ranges from 10 μm – 100 μm in diameter
Cell wall	Usually present; chemically complex in nature	When present, chemically simple in nature
Nucleus	Absent. Instead, they have a nucleoid region in the cell	Present
Ribosomes	Present. Smaller in size and spherical in shape	Present. Comparatively larger in size and linear in shape
DNA arrangement	Circular	Linear
Mitochondria	Absent	Present
Cytoplasm	Present, but cell organelles absent	Present, cell organelles present
Endoplasmic reticulum	Absent	Present

Plasmids	Present	Very rarely found in eukaryotes
Ribosome	Small ribosomes	Large ribosomes
Lysosome	Lysosomes and centrosomes are absent	Lysosomes and centrosomes are present
Cell division	Through binary fission	Through mitosis
Flagella	The flagella are smaller in size	The flagella are larger in size
Reproduction	Asexual	Both asexual and sexual
Example	Bacteria and Archaea	Plant and Animal cell

PROKARYOTIC CELL VS EUKARYOTIC CELL



Cell wall (exo-skeleton)

Cell wall is an outer rigid semi-elastic supportive and protective layer present in plant cells, bacteria, fungi and some protists.

Structure of cell wall

Cell wall is derived from the living protoplast. It consists of middle lamella, primary cell wall, plasmodesmata, secondary cell wall and pits.

1. Middle lamella: It is the first formed layer after cytokinesis which is present in between two adjacent cells.

It is made up of pectates of calcium and magnesium.

the pectin layer of middle lamella is hydrophilic colloidal substances and can absorb water and act as connecting material between two cells.

2. Primary cell wall

It is formed by protoplasm on either side of middle lamella in a growing cell.

It is chemically made up of cellulose, hemicellulose and pectic compounds.

Primary cell wall is elastic, comparatively thin and permeable.

The cells most of leaves, fruits, cortex and pith contain only the primary cell wall and middle lamella.

Secondary cell wall: The secondary cell wall is laid down on the primary cell wall when the latter has finished its growth.

It is thick ,rigid ,inelastic in nature.

It is found in certain mature and highly specialized cells such as tracheid's , vessels, collenchymatous cells, schlerenchymatous cells etc.

Chemically it is composed of lignin in addition to cellulose, hemicellulose and pectin.

Some other chemical substances like cutin, suberin, waxes, mucilage etc are deposited on it.

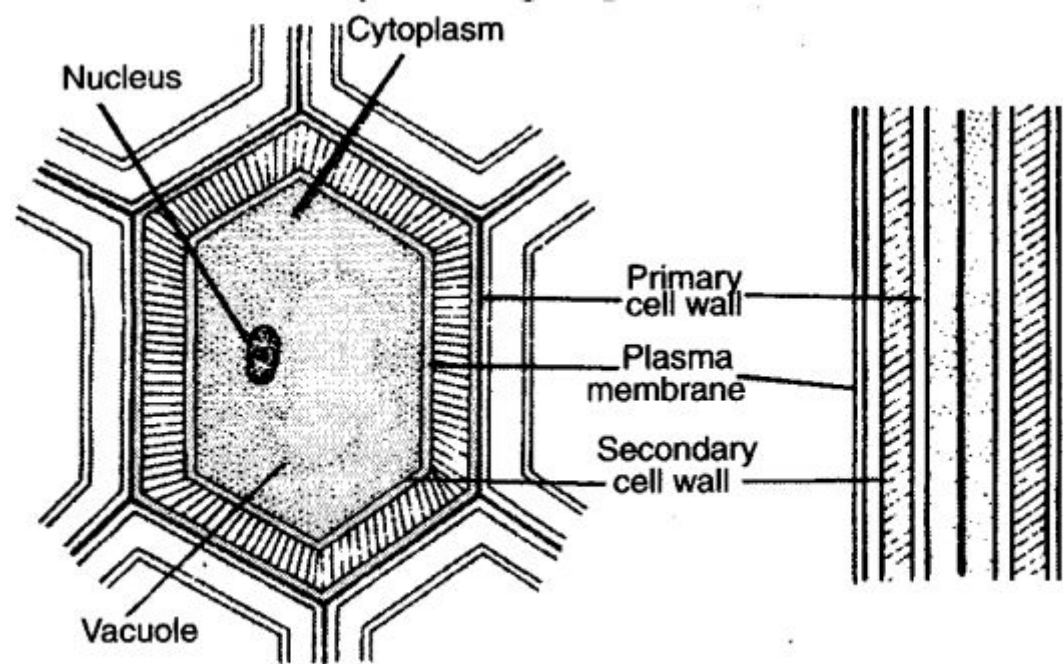


Fig. Structure of the cell wall

Plasmodesmata

The plasmodesmata were first reported by Strasburger.

The primary cell wall and middle lamella never occur in the form of continuous layers but have many minute aperture through which the cells of a tissue maintain cytoplasmic relations with each other.

Such cytoplasmic junctions or bridges between the adjacent cells are known as Plasmodesmata or pits.

Plasmodesmata permit circulation of fluid and passage of solute between cells.

Functions of cell wall

1. It provides mechanical strength to the cells and yo the plant as a whole.

2. It protects the inner contents of the cell from mechanical injuries and checks entry of germs.
3. It maintains the shape of the cell.
4. Cell wall of root hairs helps in absorption of water and minerals from the soil.
5. Suberin and cutin deposited on cell wall prevent evaporation of water.
6. It is involved in many enzymatic activities.
7. It helps in keeping a balance between intracellular osmotic pressure and that of environment and thus prevents the osmotic bursting of the cell.

Protoplasm

The term Protoplasm was coined by Johannes Purkinje. It has been defined as the Physical basis of life by Huxley because it is the fundamental substance of all organisms.

It is a living substance, since it help in metabolism.

It shows intracellular movement called Streaming movement or Cyclosis.

Chemical composition of protoplasm

Protoplasm is a very complex colloidal system of substance.

The average chemical composition of protoplasm is as follows:

Water= 75%-90%

Proteins=7%-20%

Carbohydrates=1%

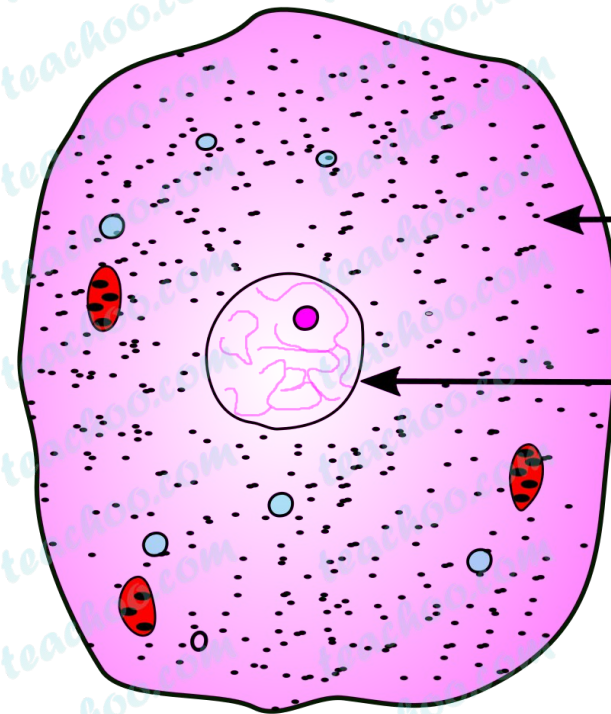
Lipids=2-3%

Inorganic materials=1%

RNA=0.7%

DNA= 0.4%

Protoplasm and Cytoplasm



Cytoplasm
+
Nucleus

Protoplasm

Major elements constituting protoplasm are

Carbon=20%

Oxygen=62%

Hydrogen=10%

Nitrogen= 3%

Trace elements (Ca ,P ,Cl ,S ,K ,Na ,Mg ,I ,Fe)

Cell membrane (biological- membrane)

All living cells are externally covered by a thin , transparent, electron microscopic, elastic, regenerative and selectively permeable membrane called plasma membrane or plasmalemma.

It was called cell membrane by Nagelli and Cramer but was called Plasmalemma by Plowe.

The plasma membrane and all the membranes covering various cell organelles present inside the cell have the same ultra structure and are collectively called bio membranes.

Biological membranes-----i. Plasma membrane

ii. Sub cellular

membranes

(cover cell organelles

within a cell)

Chemical composition of cell membrane

Cell membranes are lipoproteins i.e. they are chiefly composed of lipids and proteins. Carbohydrates are also found in the plasma membrane. The membrane contains three different types of proteins , structural proteins , enzymes (functional proteins) and carrier proteins.

Cell membranes are of two types

1. Cytoplasmic membrane which surround the whole cytoplasm.
2. Internal membrane which surrounds various cellular organelles. And vacuole.

Structure of cell membrane

Several models are there for the structure of cell membrane.

Two important models are there , they are

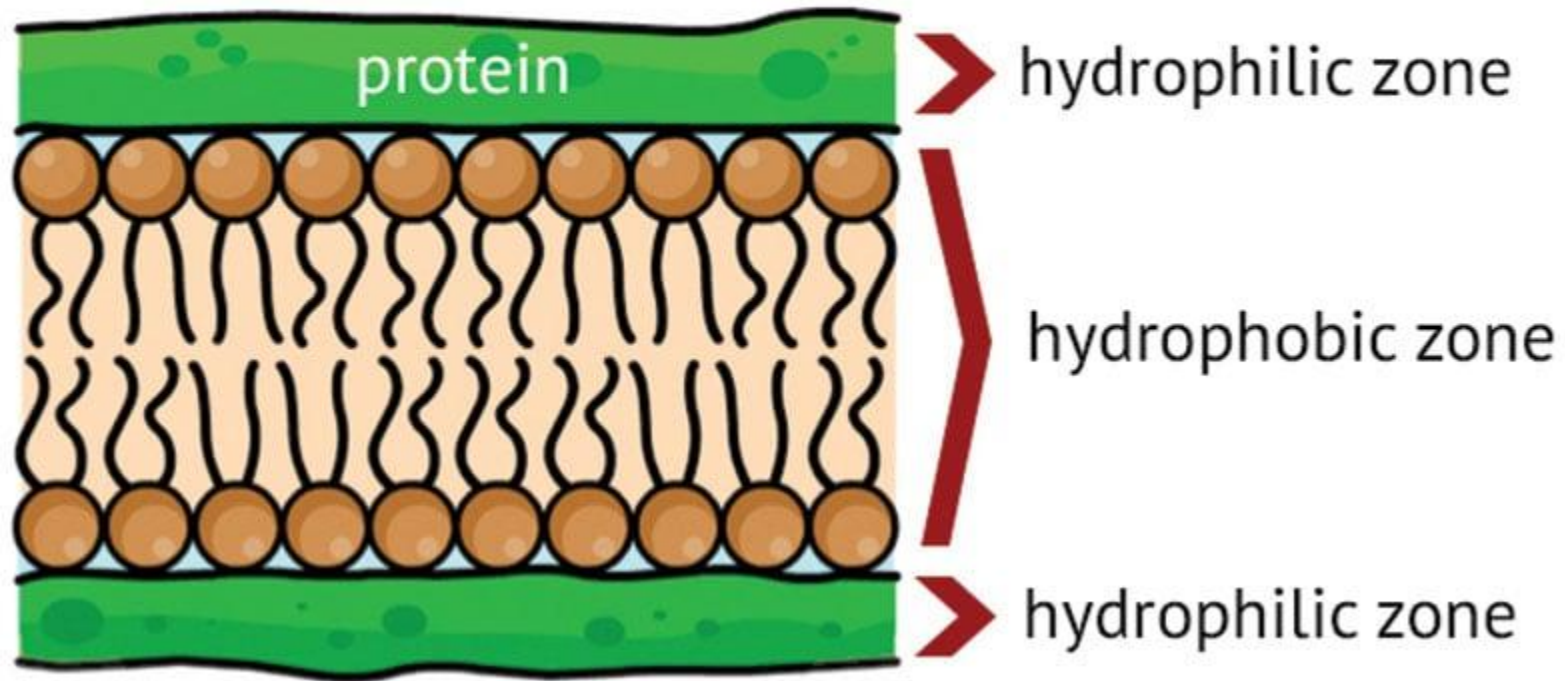
1. Sandwich model
2. Fluid mosaic model.

Sandwich model

Proposed by J. Danielli and H. Davson in 1935 A.D.

It was based on the basis of physical and chemical properties of cell membrane.

Sandwich (Davson-Danielli) model of cell membrane



It states that

1. Plasma membrane is a lipoproteinous membrane. It is formed of outer and inner denser monolayer and a middle phospholipid bilayer. So, the plasma membrane is sandwich of “protein-lipid- protein”.
2. Protein molecules are globular and give stability to the plasma membrane.
3. Each phospholipid molecule is a polar molecule and has two specific ends polar hydrophilic (water loving) head made up of glycerol and nonpolar hydrophobic (water repelling) two tails of fatty acids..
4. Hydrophilic head lies towards the periphery or towards the proteins whereas hydrophobic tails are facing towards the center.
5. Strong electrostatic force hold the hydrophilic head of phospholipid molecules together whereas the hydrophobic tails of two lipids layer are hold together by weak vander Waals forces.

Limitations of sandwich model

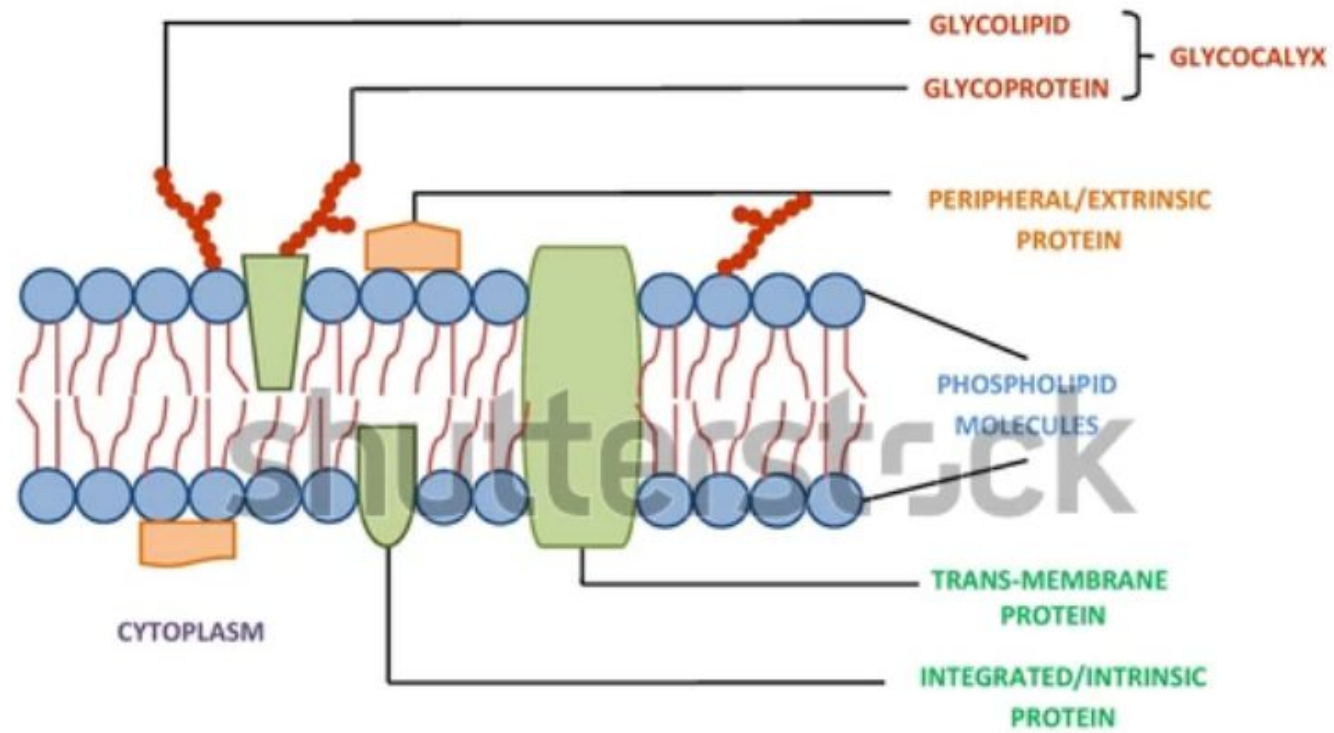
1. It does not explain the functional specificity and variability in the bio membranes.
2. It does not explain the phenomena like active transport and movement of water-soluble-substances.

Fluid- MosaicModel

This model was proposed by Singer and Nicholson in 1972 and states that “Plasma membrane is formed of protein icebergs in a sea of lipids “.

The important features of this model are as follows

1. Plasma- membrane is lipoproteinous and trilaminar.
2. Phospholipid bilayer is fluidy in nature in which phospholipid molecules show two types of movements molecules.
3. The proteins are globular.
4. Each phospholipid molecule is a polar molecule and has two specific ends : hydrophilic head of glycerol and hydrophobic tail of two fatty acid chains .So, it is an amphipathic molecule.
5. The heads of phospholipid molecules of two layers are directed in opposite directions while the tails of two layers face each other. This arrangement foms a water-resistant barrier.



FLUID MOSAIC MODEL

Functions of Plasma-membrane

1. Plasma membrane is a differentially permeable membrane which regulates flow of materials in and out of the cell . This flow may be due to concentration gradient, e.g. Diffusion, osmosis, active transport etc.
2. Plasma membrane help in endocytosis which is intake of solid or liquid materials inside the cell through plasma- membrane. It is of two types: Phagocytosis intake of solid particles called as cell eating. Pinocytosis intake of liquid nutrient called as cell-drinking.
3. Exocytosis: it is the process of expelling large sized materials outside the cell through cell membrane.
4. Cell membrane helps in cellular locomotion in two ways: Pseudopodial and undulatory.

CYTOPLASM

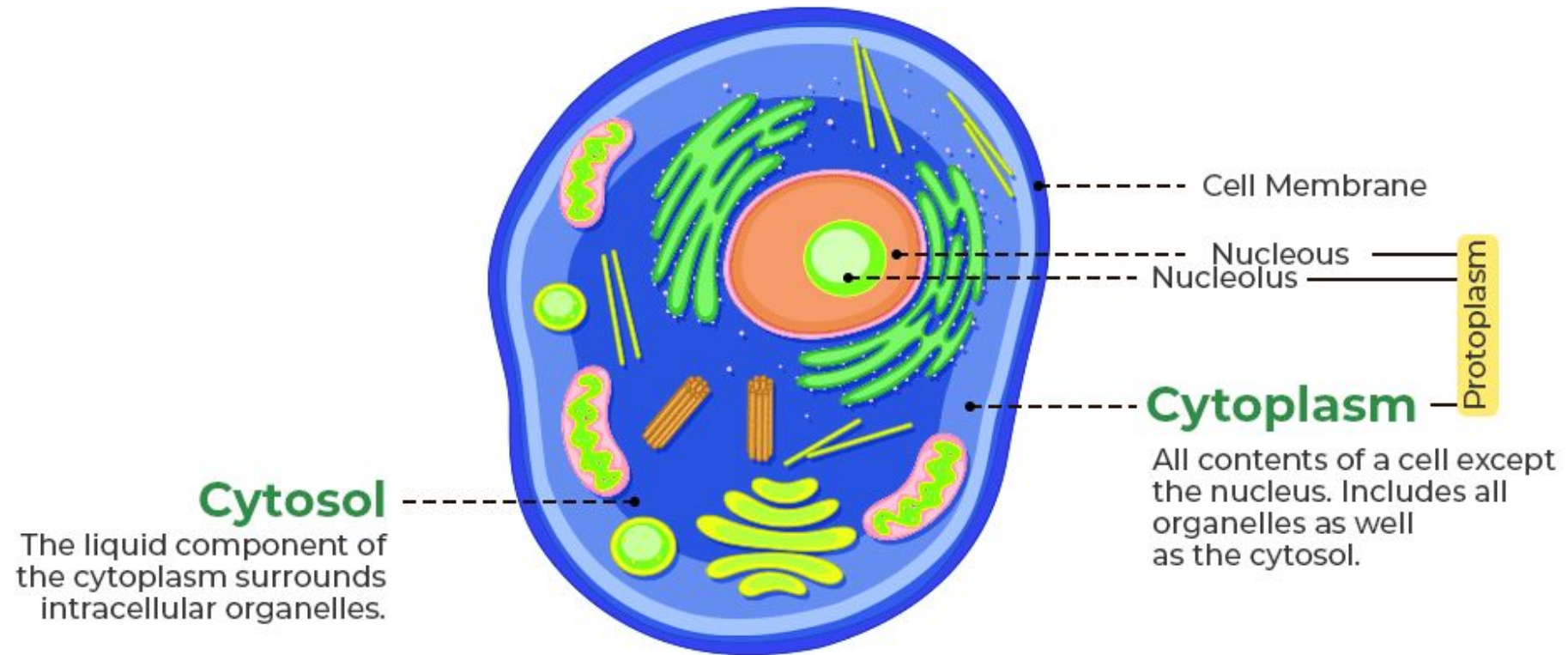
The homogeneous and colloidal ground substance lying between the plasma-membrane and nucleus is known as cytoplasm.

It is differentiated into two parts: (a) Cytoplasmic matrix
(b) Cytoplasmic structure

A. Cytoplasmic matrix

It is also known as hyaloplasm or cytosol. The cytoplasmic matrix consists of various inorganic molecules such as water, salts of Na, K and other metals and various organic compounds, like carbohydrate, lipids, proteins, nucleoproteins, nucleic acids and variety of enzymes.

The peripheral cytoplasm is called as ectoplasm or plasmagel. The inner portion is called as plasmosol or endoplasm.



Cyclosis

It is active mass movement of cytoplasm around the central vacuole in one direction. It is also known as cytoplasmic streaming.

Cytoplasmic structures(cell organelles and cell inclusions)

Cell organelles They are living sub –cellular structures of the cytoplasm.

Mitochondria

Mitochondria were first discovered by Kolliker in the voluntary muscle of insects.

The mitochondria term was given by C. Benda.

Mitochondria are granular structure and are also known as “Power house of the cell” as they release energy in the Form of ATP during aerobic respiration for the vital activities of a cell.

Occurance

They are absent from prokaryotes but present in all eukaryotic cells except mature and old mammalian RBC and sieve tube of phloem in vascular plants. Mitochondria are distributed throughout the cytoplasm.

Shape

Mitochondria may be rod shaped, filamentous, spherical, oval shaped and may change from one form to another .

Size

Their size varies from cell to cell. Smallest sized mitochondria are found in yeast cells and largest sized mitochondria are found in toad.

They are largest organelles in animal cell and second largest organelle in plant cell and third largest structure in the cell after nucleus and chloroplast.

Number

Their number varies and depends on the type and functional state of the cell.

Chlamydomonas - one (1)

Muscle cell of insect -500000

Position

Mitochondria are generally located in the metabolically active area of the cells to provide ATP for quick cellular activities.

Ultrastructure

Mitochondria are bound by two membranes, the outer membrane and the inner membrane. The space between the two membranes is called outer chamber which is filled with a watery fluid.

The space enclosed by the inner membrane is called inner chamber. It is filled with a matrix which contains dense granules of insoluble inorganic salts, ribosomes and mitochondrial DNA. The enzymes of Krebs's cycle are located in matrix.

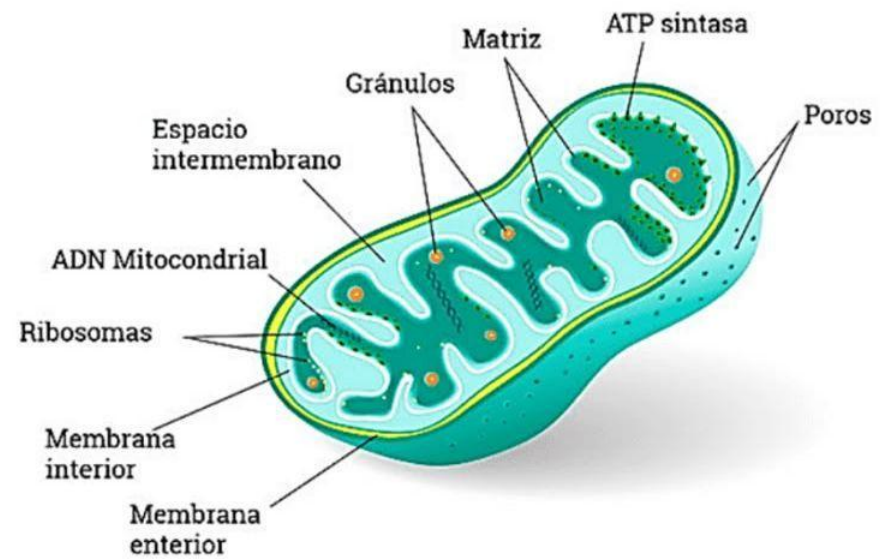
The outer membrane is smooth while the inner membrane is enfolded to form Cristae which increase surface area of cell respiration.

On the matrix side of inner membrane several thousands of tennis racket like structures are present known as elementary particles or F1- particles or oxysomes .

Each elementary particle consists of a head piece, stalk and base. Elementary particles are involved in the process of oxidation and phosphorylation reactions.

In the head of F1 particle, enzymes ATPase or ATP synthetase is present which controls synthesis of ATP so these are also called ATP particles and are the functional units of mitochondria.

MITOCONDRIA



Composition

Chemical composition of mitochondria varies in different animals and plant cells; however, they are found to contain protein, lipids, RNA and small amount of DNA.

Semi autonomous nature of mitochondria

Mitochondria is a semi-autonomous cell organelle because 70s ribosome, self replicating DNA and has all requirements of protein synthesis.

Functions

1. Mitochondria are known as powerhouse of the or storage batteries or ATP factories as these are sites of ATP formation.
2. Intermediate products of cellular respiration are used in the formation of steroids, cytochromes ,chlorophylls etc.
3. These are also site of biosynthesis of some amino acids.
4. Mitochondria also regulate the calcium ion concentration inside the cell.
5. The help in yolk formation during the development of ovum.
6. They form middle piece of sperm during sperm maturation.
7. They are the site of synthesis of haeme of haemoglobin and myoglobin.
8. They are also the site of thermogenesis.

PLASTIDS

Plastids are the double membrane bounded cytoplasmic organelles of the cell of plants and some protozoans such as Euglena.

They develop from small bodies called proplastids found in the meristematic regions.

They are absent in bacteria, blue green algae, fungi and animals which contain chromatophores instead of plastids.

They may be colorless or colored.

One form of plastids can change into another form.

For example, in tomato, the young ovary contains leucoplasts which change into chloroplasts in young fruits and finally into chromoplasts as the fruit ripens.

The term plastid was given by Haeckel.

Schimper divided plastids in three types.

Occurrence

These are found in all plant cells and some protists but are absent from Monera, fungi and animalia. They are absent from gametes and zoospores of plants.

Types

Chloroplast

The most common of the plastids are chloroplasts. They are very important for plants, because photosynthesis by which complex organic food is manufactured, takes place in them.

Chloroplasts were first observed by Anton Von Leeuwenhock. The term chloroplasts was given by Schimper. These are green plastids possessing green pigment called chlorophylls.

occurrence

They are most common type of plastids found in all photosynthetic cells of plants. They are also found in some protists like Euglena. These are absent in those parts of plants which are not exposed to light.

Shape

Their shape varies widely. In higher plants chloroplasts are generally biconvex or plano-convex.

However, in different plant cells, it may have various shapes such as filamentous , saucer shaped, ovoid, discoid , spheroid, star-shaped, spiral ribbon like, reticulate or cup-shaped.

Size

The size of chloroplasts also varies from species to species; chloroplasts generally measures 2–3-micron meter in thickness and 5-10 micron in diameter.

Number

It varies from cell to cell and depends on the physiological state of cell. E.g. Only one chloroplast in Chlamydomonas, 1-16 in spirogyra, 500 per cell in internodal cell of Chara.

Ultrastructure

Chloroplasts are bounded by two-unit membranes, an outer and an inner lipoproteinous membranes with an inter membrane space between them.

The inner membrane encloses Stroma or Matrix which is a denser, colorless and granular ground substances.

The stroma is mainly formed of proteins, circular and necked DNA molecules, mRNA, and tRNA molecules, water, minerals and enzymes. It the site for dark reaction of photosynthesis.

In the stroma, small structures called Grana and intergrana connecting membranes are present.

Each granum consist of disc- shaped membranous sacs called thylakoids piled one upon others.

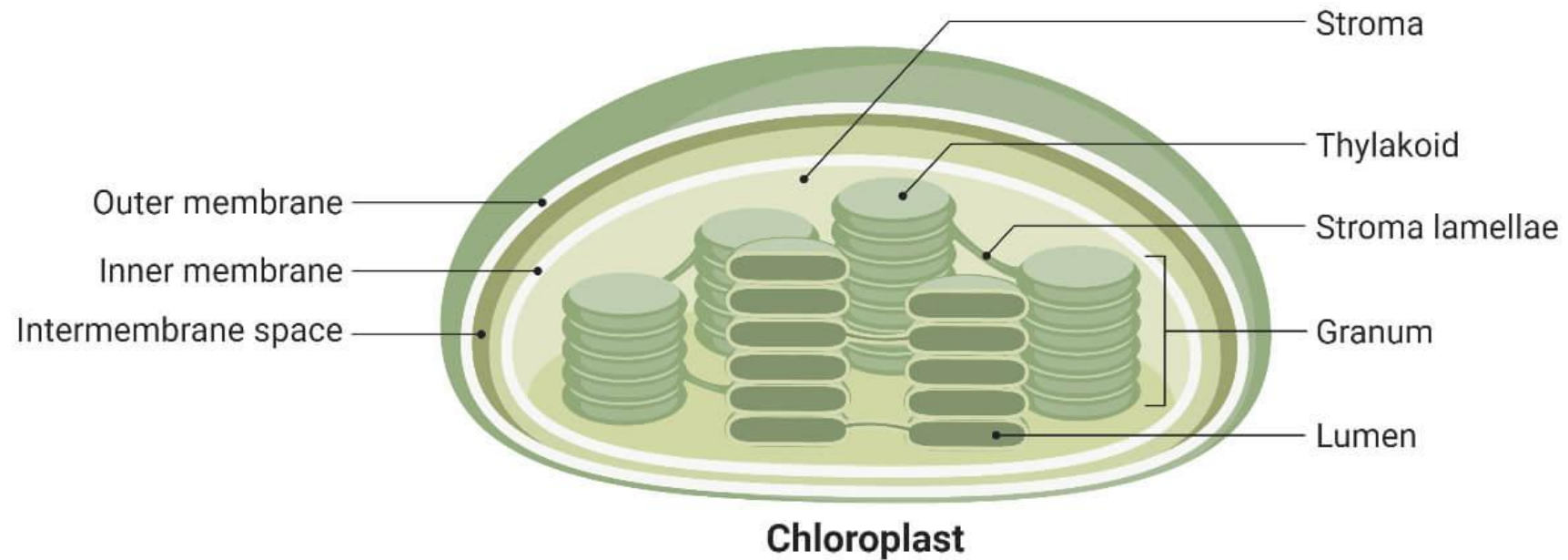
Grana are interconnected by a network of membranous tubules known as the intergrana or stroma Lamellae frets.

The thylakoid membrane contains all the enzymatic components required for photosynthesis.

They are the site of light reactions . Interaction between chlorophyll and other components takes place within the thylakoid membrane.

Chloroplast

Definition, Structure, Types, Functions, Diagram



Semi autonomous nature of chloroplasts

Like mitochondria, chloroplasts are semi-autonomous cell organelles because they have a complete machinery to synthesize some of the required proteins while for some other proteins these depend upon nuclear DNA and cytoplasmic ribosomes.

Functions

1. The most important and fundamental function of chloroplast is photosynthesis. Chloroplasts are the kitchens of cells.
2. They evolve oxygen for respiration for all the aerobes.
3. They maintain O_2 and CO_2 balance in the biosphere.
4. They prevent global warming by reducing CO_2 concentration.

5. They store the starch in the proteinaceous bodies called pyrenoids in algal forms.
6. In spinach, synthesis of fatty acids occurs in chloroplasts.
7. They help in formation of ATP and NADPH₂ during light reaction of photosynthesis which is utilized in dark reaction of photosynthesis.
8. Chloroplast store starch temporarily in higher plants.
9. Chloroplasts store fats as plastoglobuli which are rich in vitamin K and E.

Chromoplasts

Chloroplasts are non-photosynthetic colored plastids which synthesize and store carotenoid pigments. So, they appear orange, red or yellow in color.

Chromoplasts are developing from proplastids, chloroplast and leucoplast.

Chloroplast can be changed into chromoplasts during ripening of fruits like chili, tomato, etc.

Colored petals are formed by changing proplastids into chromoplasts.

Color of carrot root is due to formation of chromoplast from leucoplasts.